Project Check Point 1

Team Name: S.E.N.D.
Project Title: Marketplace Product Price Comparator App
Course: CIS 553 - 002, Fall 2022
Group Members: Nayan Patel, Steve Anderau, Ethan Pellegrini, Dheeraj Mohanbabu

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- What functionality does the system have at the end of this iteration?

ANS: With this iteration (i.e., Sprint 2), team is expecting to have framework ready for frontend and backend and also, the framework for the data exchange between those is working.

Following system functionality has been achieved at the end of this iteration:

1) **The framework for the frontend (user interface (UI)) is ready with some basic functionality:**
    a. **Search text box**
    b. **Search start button**
2) **The framework for backend (webscraping module) is working for scraping product data from amazon.com and bestbuy.com marketplace sites.**
3) **The framework for data exchange between frontend and backend is working and proven with basic testing.**

- List the user stories that you have implemented at this iteration.

ANS: Following stories/tasks that we have implemented in this iteration (i.e., Sprint 2):

Table 1							
Issue Type	**Issue key**	**Issue id**	**Summary**	**Assignee**	**Priority**	**Status**	**Description**

Story	SEND-31	10030	Framework for data exchange between backend (webscraping) and frontend (UI)	Nayan Patel	Medium	In Progress	As a user of the "Marketplace Product Price Comparator App", I want to be able to see a framework is available where the data which are entered in frontend (Graphical User Interface (GUI)) are pushed to the backend (webscraping module). Also, I want to see the data that is generated by the backend is sent to/fetched by the frontend. Pre-conditions: * Framework for UI is available or a sample UI code is available to test Post-conditions: Data between frontend and backend are exchanged without any issue

Task	SEND-24	10023	Create flexibility for adding marketplace sites in webscraping code	dheerajm	Medium	In Progress	Using framework, add flexibility (e.g., class, methods etc.) that can be used to add more marketplace sites PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product POST-CONDITIONS: The webscraping framework can add as many as needed marketplace sites search a product across all the sites and get desired result

Task	SEND-23	10022	Create a framework for webscraping a product from a marketplace site	Steve Anderau	Medium	In Progress	Create a framework in Python using Beautifulsoup and/or Selenium libraries to scape a marketplace site for searched product's attributes. PRE-CONDITIONS: * The libraries to use for webscraping is identified and installed * The investigation on what the product attributes to scrape from all the marketplace sites is complete POST-CONDITIONS: The framework for the webscraping is available with following features: # It can successfully load the marketplace site # It can search a product

Task	SEND-21	10020	Create a User Interface framework in Python	Ethan Pellegrini	Medium	In Progress	Create a user interface framework in Python PRE-CONDITIONS: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed POST-CONDITIONS: The UI framework is working and tested with some simple user interfaces (e.g., a box, button etc.)

- Did you end up making any changes to any of these user stories? Did you break down further any the user stories? Did you identify any new user stories during this iteration and, if so, did you add them to the product backlog or decide to implement them right away? Explain.

ANS: Our team did not make any changes to the originally planned user stories for this iteration nor we needed to further break down the stories. However, based on the need, we added following new user story to this iteration:

Table 2							
Issue Type	**Issue key**	**Issue id**	**Summary**	**Assignee**	**Priority**	**Status**	**Description**

Story	SEND-31	10030	Framework for data exchange between backend (webscraping) and frontend (UI)	Nayan Patel	Medium	In Progress	As a user of the "Marketplace Product Price Comparator App", I want to be able to see a framework is available where the data which are entered in frontend (Graphical User Interface (GUI)) are pushed to the backend (webscraping module). Also, I want to see the data that is generated by the backend is sent to/fetched by the frontend. Pre-conditions: * Framework for UI is available or a sample UI code is available to test Post-conditions: Data between frontend and backend are exchanged without any issue

- What are the "lessons learned" at the end of this iteration? What would you do differently next time? Explain.

ANS: The SEND team has learned some lessons from this iteration and those are going to be used to improve the practice in next iterations.

1. **Initially, research was done to see if any marketplace websites have an open API, making it more efficient and robust to extract marketplace product information. After multiple attempts, it was found that these marketplace websites had API's but were either behind a paywall or for specific users only. The lesson learned for this situation is to supply more focus and resources to the use case at hand rather than spending more time focusing on a variation of the use case that has a smaller percentage of working.**

2. **During web scraping development it was found that not all of the product information needed to be extracted could be done so with pure html parsing. Certain product information could only be found in the html post javascript rendering using a headless browser system with the package Selenium. As this is a harder error to diagnose, there were multiple testing iterations using pure html parsing before the solution became apparent. The lesson learned from this situation is that sometimes it can be good to step away from a problem and perform some research on it before getting back into it.**
3. **How to properly research building an application from the ground up**
4. **We learned that the first step to any good UI is a good design and without having a good design upfront, it would be lots of rework**
5. **Properly weighting our stories really helped to divvy the work well (we did this well, but may not hurt to have as an observation)**
6. **Building a UI is often a much harder task than it appears, requiring much iteration. Stories should be broken down to really reflect that if building a UI is the goal**

- Provide an updated numbered list of all user stories yet to be implemented; indicate pre- and post-conditions.

ANS: Following stories/tasks are yet to be implemented and they have been assigned to two remaining sprints, i.e., Sprint 3 and 4

Table 3							
Issue Type	**Issue key**	**Issue id**	**Summary**	**Assignee**	**Priority**	**Status**	**Description**

Task	SEND-29	10028	Display sorted results on the UI	Medium	To Do	Pull the sorted results from the data structure and display on the UI Pre-conditions: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure Post-conditions: The sorted table from the data structure shall be displayed on the UI
Task	SEND-28	10027	Sort the result in the background	Medium	To Do	Sort the result table in the background and store in the same or a different data structure Pre-conditions: The user selection for the sorting by product attribute is available Post-conditions: Based on the user selected product attribute for sorting, the result table shall be sorted and the sorted table is stored in the same or a new data structure

Task	SEND-27	10026	Create an option in UI to sort results	Medium	To Do	Create an option in UI that can allow the user to choose the attribute that is used to sort the results Pre-conditions: The drop down (or similar) menu in the GUI is available to select the attribute that user would like to sort the list by Post-conditions: The user selected product attribute for sorting shall be captured in a UI backend variable with correct value

Task	SEND-26	10025	Result a result table in a data structure form	Medium	To Do	The product search result (see following) shall be stored in a data structure that can be used by the UI to display the results: * Product name * Marketplace site name * Product Attribute 1 * Product Attribute 2¦¦ * Product Attribute n PRE-CONDITIONS: The webscraping code shall return all desired product attributes POST-CONDITIONS: The product details along with the product attributes shall be returned in a table (data structure) form

Task	SEND-25	10024	Add functionality to webscraping framework to read product attributes	Medium	To Do	Add code to read out the product attributes to previously created webscraping framework. PRE-CONDITIONS: The framework for the webscraping is available with following features: * It can successfully load the marketplace site * It can search a product * It search products in several different market place sites POST-CONDITIONS: The webscraping code shall return all desired product attributes

Task	SEND-22	10021	Add user selection options in the user interface	Medium	To Do	Add following options in the user interface framework: # Text box to enter the keyword for searching the product # Select all the marketplace sites by clicking on the checkboxes against their names # Click button to start search PRE-CONDITIONS: The framework for the UI is ready and working without any error. POST-CONDITIONS: The options listed above shall be available and working without any problem.

Story	SEND-20	10019	Integration Testing	Medium	To Do	As a user of the Marketplace Product Price Comparator App, I want to run an end to end integration testing for all the major use cases: # Search for the product # Display search results # Sort search results by attribute Pre-conditions: The stories SEND-17, SEND-18, and SEND-19 are complete and unit testing was done for those Post-conditions: All the integration tests passed

Story	SEND-19	10018	Develop a User Interface for the App to sort the search results	Medium	To Do	As a user of the Marketplace Product Price Comparator App, I want to be able to sort the search results for attributes of a product from various marketplace sites using a option through a Graphical User Interface (GUI), so that I can get list sorted by desired attribute (e.g., price, ratings etc.). Pre-conditions: The drop down menu in the GUI is available to select the attribute that user would like to sort the list by Post-conditions: Customer receives the sorted compared list of attributes of the desired product

Story	SEND-18	10017	Develop web scraper module to search and display product attributes on GUI	Medium	To Do	As a user of the Marketplace Product Price Comparator App, I want to be able to see the search result list (product attributes) from different marketplace sites on a Graphical User Interface (GUI) , so that I can compare them to find the best match (e.g., best price, best rating, etc.). Pre-conditions: * The libraries to use for webscraping is identified and installed * The investigation on what the product attributes to scrape from all the marketplace sites is complete Post-conditions: The list of product attributes along with the names of the marketplace sites (un-sorted list) is displayed on the GUI

Story	SEND-17	10016	Develop a User Interface for the App for searching product by keyword	Medium	To Do	As a user of the Marketplace Product Price Comparator App, I want to be able to search the product using a keyword through a Graphical User Interface (GUI), so that I can get the price and other attributes of the product from different marketplace sites . Pre-conditions: # The set up for Python code development (IDE, interpreter etc.) is ready # The required Python libraries have been identified and installed Post-conditions: The options listed above shall be available and working without any problem.

- If this is not the last iteration: Given the current functionality of the system and considering the pre- and post-conditions, identify a subset of user stories to be implemented during the next iteration. Be sure that the cumulative size of the selected user stories is about 1/3 of the size of the full backlog. Describe the functionality that your (partially implemented) system will have at the end of this iteration.

ANS: Stories SEND-22, SEND-25 and SEND-26 from Table 3 above have been assigned to the next iteration (i.e., Sprint 3) and the total size of the stories is approx, 1/3 of the total size of the full backlog.

The expected functionality at the end of the next iteration will be:

- **All the user selection options (e.g., text box to enter keyword/product id search, selection of marketplace sites to search etc.) are available and functioning**
- **The backend (webscraping module) has functionality working and completely tested to collect product attributes (i.e., price, rating etc.)**
- **The collected searched results (i.e., product attributes) are captured in a table and can be displayed on the UI**

❖ If this is the last iteration: Are there any user stories left unimplemented in the backlog? Are there any new user stories that you would consider adding to the backlog. List these user stories and explain them.

ANS: Not Applicable as this is not the last iteration.

❖ Checklist for this deliverable:

➢ Estimated number of hours by each member:

Hours Spent on	Name of the members			
	Dheeraj Mohanbabu	Steve Anderau	Nayan Patel	Ethan Pellegrini
Week of 10/24	4	4	2	3
Week of 10/31	5	6	5.5	4
On 11/06	4	2	3.5	5
Total Hours	13	12	11	12

➢ Meeting dates since the last project deliverable:

11/02 at 12:00 PM

11/06 at 4:00 PM-5:00 PM